

Math Olympiad Trainer Academy

AY 25/26 — Cycle III | Exercise Collection

Part I — Applications of the Cauchy-Bunyakovsky-Schwarz Inequality

Warm-up Problems

Exercise 1

Let $a, b, c > 0$. Prove Nesbitt's inequality

$$\frac{a}{b+c} + \frac{b}{c+a} + \frac{c}{a+b} \geq \frac{3}{2}.$$

Exercise 2

Let $a, b, c, d \in \mathbb{R}^+$. Prove the inequality

$$\frac{c}{a+3b} + \frac{d}{b+3c} + \frac{a}{c+3d} + \frac{b}{d+3a} \geq 1.$$

Exercise 3

Let $x, y, z > 0$ be real numbers. Prove the inequality

$$\frac{x}{x+2y+3z} + \frac{y}{y+2z+3x} + \frac{z}{z+2x+3y} \geq \frac{1}{2}.$$

Exercise 4

Let $a, b \in \mathbb{R}^+$ be such that $ab \geq 1$. Prove the inequality

$$\left(a + 2b + \frac{2}{a+1}\right) \left(b + 2a + \frac{2}{b+1}\right) \geq 16.$$

Exercise 5

Let $a, b, c, d > 1$ be real numbers. Prove the inequality

$$\sqrt{a-1} + \sqrt{b-1} + \sqrt{c-1} + \sqrt{d-1} \leq \sqrt{(ab+1)(cd+1)}.$$

Exercise 6

Let a, b, c be positive real numbers such that $a+b+c=1$. Prove the inequality

$$a\sqrt{b} + b\sqrt{c} + c\sqrt{a} \leq \frac{1}{\sqrt{3}}.$$

Intermediate Problems

Exercise 7

Let $a, b, c \in (0, 1)$ be real numbers. Prove the inequality

$$\sqrt{abc} + \sqrt{(1-a)(1-b)(1-c)} < 1.$$

Exercise 8

Let $a, b, c > 0$ be real numbers such that

$$\frac{1}{a} + \frac{1}{b} + \frac{1}{c} = 2.$$

Prove the inequality

$$\sqrt{a-1} + \sqrt{b-1} + \sqrt{c-1} \leq \sqrt{a+b+c}.$$

Exercise 9

Let a, b, c be positive real numbers. Determine the minimal value of

$$\frac{3a}{b+c} + \frac{4b}{c+a} + \frac{5c}{a+b}.$$

Exercise 10

Let a, b, c be positive real numbers such that $ab + bc + ca = 1/3$. Prove the inequality

$$\frac{a}{a^2 - bc + 1} + \frac{b}{b^2 - ca + 1} + \frac{c}{c^2 - ab + 1} \geq \frac{1}{a+b+c}.$$

Exercise 11

Let $a, b, c \in \mathbb{R}^+$ such that $abc = 1$. Prove the inequality

$$\frac{a}{1+bc} + \frac{b}{1+ca} + \frac{c}{1+ab} \geq \frac{3}{2}.$$

Exercise 12

Let a, b, c be positive real numbers such that $a + b + c = 3$. Prove the inequality

$$\frac{a^3+2}{b+2} + \frac{b^3+2}{c+2} + \frac{c^3+2}{a+2} \geq 3.$$

Exercise 13

Let $a, b, c \in \mathbb{R}^+$ such that $a^2 + b^2 + c^2 = 1$. Prove the inequality

$$a + b + c - 2abc \leq \sqrt{2}.$$

Exercise 14

Let a, b, c, d be positive real numbers such that $abcd = 1$. Prove the inequality

$$(1+a^2)(1+b^2)(1+c^2)(1+d^2) \geq (a+b+c+d)^2.$$

Challenges**Exercise 15**

Let $a, b, c \geq 1$ be real numbers. Prove the inequality

$$\sqrt{a-1} + \sqrt{b-1} + \sqrt{c-1} \leq \sqrt{a(bc+1)}.$$

Exercise 16

Let $a, b > 0$ be such that $ab \geq 1$. Prove the inequality

$$\left(\frac{a^3}{a+1} + b + 1 \right) \left(\frac{b^3}{a+1} + a + 1 \right) \geq \frac{25}{4}.$$

Exercise 17

Let $a, b, c > 0$ be real numbers such that $abc = 1$. Prove the inequality

$$\sqrt{2(1+a^2)(1+b^2)(1+c^2)} \geq 1+a+b+c.$$

Exercise 18

Let $x, y, z \in \mathbb{R}^+$ such that $x^2 + y^2 + z^2 = 2$. Prove the inequality

$$x + y + z \leq 2 + xyz.$$

Exercise 19

Let a, b, c, d, e, f be positive real numbers. Prove the inequality

$$\frac{a}{b+c} + \frac{b}{c+d} + \frac{c}{d+e} + \frac{d}{e+f} + \frac{e}{f+a} + \frac{f}{a+b} \geq 3.$$

Exercise 20

Let a, b, c be positive real numbers such that $ab + bc + ca = 1$. Prove the inequality

$$\frac{1}{4a^2 - bc + 1} + \frac{1}{4b^2 - ca + 1} + \frac{1}{4c^2 - ab + 1} \geq \frac{3}{2}.$$

Part II — Reverse Techniques and Further Applications

Warm-up Problems

Exercise 21

Let $a, b, c > 0$ be real numbers such that $abc = 1$. Prove the inequality

$$\frac{1}{a^3(b+c)} + \frac{1}{b^3(c+a)} + \frac{1}{c^3(a+b)} \geq \frac{3}{2}.$$

Exercise 22

Let $x, y, z > -1$ be real numbers. Prove the inequality

$$\frac{1+x^2}{1+y+z^2} + \frac{1+y^2}{1+z+x^2} + \frac{1+z^2}{1+x+y^2} \geq 2.$$

Exercise 23

Let $a, b, c > 0$ be real numbers such that $a^2 + b^2 + c^2 = 3$. Prove the inequality

$$\frac{a(a-b^2)}{a+b^2} + \frac{b(b-c^2)}{b+c^2} + \frac{c(c-a^2)}{c+a^2} \geq 0.$$

Exercise 24

Let $a, b, c > 0$ be real numbers such that $abc = a + b + c + 2$. Prove the inequality

$$2(\sqrt{ab} + \sqrt{bc} + \sqrt{ca}) \leq a + b + c + 6.$$

Exercise 25

Let a, b, c and x, y, z be positive real numbers. Prove the inequality

$$a(y+z) + b(z+x) + c(x+y) \geq 2\sqrt{(xy+yz+zx)(ab+bc+ca)}.$$

Problems**Exercise 26**

Let a, b, c be positive real numbers such that $abc = 1$. Prove the inequality

$$\frac{a^2}{\sqrt{a+b}} + \frac{b^2}{\sqrt{b+c}} + \frac{c^2}{\sqrt{c+a}} \geq \frac{3}{\sqrt{2}}.$$

Exercise 27

Let $a, b, c \in \mathbb{R}^+$ be such that $abc = 1$. Prove the inequality

$$\frac{a+b}{\sqrt{b+c+2}} + \frac{b+c}{\sqrt{c+a+2}} + \frac{c+a}{\sqrt{a+b+2}} \geq 3.$$

Exercise 28

Let $a, b, c \in \mathbb{R}^+$ be such that $abc = 1$. Prove the inequality

$$\frac{a}{\sqrt[3]{a+b+1}} + \frac{b}{\sqrt[3]{b+c+1}} + \frac{c}{\sqrt[3]{c+a+1}} \geq \sqrt[3]{9}.$$

Exercise 29

Let $a, b, c > 0$ be real numbers. Prove the inequality

$$\frac{a}{2a+b} + \frac{b}{2b+c} + \frac{c}{2c+a} \leq 1.$$

Exercise 30

Let a, b, c be positive real numbers such that $a^2 + b^2 + c^2 = 3$. Prove the inequality

$$\frac{1}{2-a} + \frac{1}{2-b} + \frac{1}{2-c} \geq 3.$$

Exercise 31

Let a, b, c be positive real numbers such that $a^2 + b^2 + c^2 = 3$. Prove the inequality

$$\frac{a}{a^2+2b+3} + \frac{b}{b^2+2c+3} + \frac{c}{c^2+2a+3} \leq \frac{1}{2}.$$

Exercise 32

Let a, b, c be positive real numbers such that $abc = 8$. Prove the inequality

$$\frac{a-2}{a+1} + \frac{b-2}{b+1} + \frac{c-2}{c+1} \leq 0.$$

Exercise 33

Let $a, b, c \in \mathbb{R}^+$ such that

$$\frac{1}{a+b+1} + \frac{1}{b+c+1} + \frac{1}{c+a+1} \geq 1.$$

Prove the inequality

$$a+b+c \geq ab+bc+ca.$$

Exercise 34

Let a, b, c be positive real numbers such that

$$\frac{1}{a^2+b^2+1} + \frac{1}{b^2+c^2+1} + \frac{1}{c^2+a^2+1} \geq 1.$$

Prove the inequality

$$ab+bc+ca \leq 3.$$

Exercise 35

Let a, b, c be positive real numbers such that $abc = 1$. Prove the inequality

$$\frac{a}{a+b^4+c^4} + \frac{b}{b+c^4+a^4} + \frac{c}{c+a^4+b^4} \leq 1.$$

Exercise 36

Let $x, y, z \in \mathbb{R}^+$ be such that $x+y+z=3$. Prove the inequality

$$\frac{xy}{x+2y+3xy} + \frac{yz}{y+2z+3yz} + \frac{zx}{z+2x+3zx} \leq \frac{1}{2}.$$

Exercise 37

Let $x, y, z \in \mathbb{R}^+$ be such that $x+y+z=3$. Prove the inequality

$$\frac{xy}{x+2y+1} + \frac{yz}{y+2z+1} + \frac{zx}{z+2x+1} \leq \frac{3}{4}.$$

Exercise 38

Let $x, y, z \geq 0$ be real numbers. Prove the inequality

$$\sqrt{x^2+1} + \sqrt{y^2+1} + \sqrt{z^2+1} \geq \sqrt{6(x+y+z)}.$$

Exercise 39

Let $x, y, z > 0$ be real numbers such that $x+y+z=3$. Prove the inequality

$$\sqrt{x^3+x^2+x+7} + \sqrt{y^3+y^2+y+7} + \sqrt{z^3+z^2+z+7} \geq 3\sqrt{10}.$$

Exercise 40

Let x, y, z be positive real numbers. Prove the inequality

$$xy(x+y-z) + yz(y+z-x) + zx(z+x-y) \geq \sqrt{3(x^3y^3+y^3z^3+z^3x^3)}.$$

Exercise 41

Let a, b, c be positive real numbers such that $ab + bc + ca = 3$. Prove the inequality

$$\frac{a(b^2 + c^2)}{a^2 + bc} + \frac{b(c^2 + a^2)}{b^2 + ca} + \frac{c(a^2 + b^2)}{c^2 + ab} \geq 3.$$

Challenges**Exercise 42**

Let $x, y, z \in [-1, 1]$ be real numbers such that $x + y + z + xyz = 0$. Prove the inequality

$$\sqrt{x+1} + \sqrt{y+1} + \sqrt{z+1} \leq 3.$$

Exercise 43

Let $x \geq y \geq z > 0$ be real numbers. Prove the inequality

$$\frac{x^2 y}{z} + \frac{y^2 z}{x} + \frac{z^2 x}{y} \geq x^2 + y^2 + z^2.$$

Exercise 44

Let a, b, c be positive real numbers such that $a + b + c = 1$. Prove the inequality

$$(a+b)^2(1+2c)(2a+3c)(2b+3c) \geq 54abc.$$

Exercise 45

Let $a, b, c > 0$ be real numbers such that $a + b + c = 3$. Prove the inequality

$$\frac{1}{a^2 + b^2 + 2} + \frac{1}{b^2 + c^2 + 2} + \frac{1}{c^2 + a^2 + 2} \leq \frac{3}{4}.$$

Exercise 46

Let $a, b, c > 0$ be real numbers such that $abc = 1$. Prove the inequality

$$\frac{a}{\sqrt{b+c+2}} + \frac{b}{\sqrt{c+a+2}} + \frac{c}{\sqrt{a+b+2}} \geq \frac{3}{2}.$$

Exercise 47

Let a, b, c be positive real numbers such that $abc = 1$. Prove the inequality

$$\frac{1}{2+a} + \frac{1}{2+b} + \frac{1}{2+c} \leq 1.$$

Exercise 48

Let x, y, z be positive real numbers such that $xyz \geq 1$. Prove the inequality

$$\frac{x^5 - x^2}{x^5 + y^2 + z^2} + \frac{y^5 - y^2}{y^5 + z^2 + x^2} + \frac{z^5 - z^2}{z^5 + x^2 + y^2} \geq 0.$$

Exercise 49

Let a, b, c be positive real numbers such that $abc \geq 1$. Prove the inequality

$$\frac{1}{a^4 + b^3 + c^2} + \frac{1}{b^4 + c^3 + a^2} + \frac{1}{c^4 + a^3 + b^2} \leq 1.$$

Exercise 50

Let $x, y, z \in \mathbb{R}^+$ be such that $x + y + z = 3$. Prove the inequality

$$\frac{1}{x + 2yz + 3zx} + \frac{1}{y + 2zx + 3xy} + \frac{1}{z + 2xy + 3yz} \leq \frac{1}{2xyz}.$$

Exercise 51

Let a, b, c be positive real numbers. Prove the inequality

$$ab^{\frac{a+c}{b+c}} + bc^{\frac{b+a}{c+a}} + ca^{\frac{c+b}{a+b}} \geq \sqrt{3abc(a+b+c)}.$$